Table of Contents

Contents

Data Types:....2

Business Logic Constraints: 3

Task Decomposition with Abstract Code:....4

Login....4

Register 5

View main menu....6

List Item to Sell....7

Item Search Query....8

View Description & Bids....9

Edit Description....10

Cancel Auction....12

View Ratings....13

Delete Ratings....14

Auction Results 14

Item Results & Winners 15

Submit Ratings 16

Report Queries 17

- Category Report 17
- User Report....18
- Top Rated Items Report 18
- Auction Statistics 19
- Canceled Auction Details 20

Data Types:

User

Attribute	Data type	Nullable
first_name	String	Not Null
last_name	String	Not Null
user_name	String	Not Null
password	String	Not Null
position	String	Null

Item

Attribute	Data type	Nullable
item_id	Integer	Not Null
item_name	String	Not Null
description	String	Not Null
returnable	Boolean	Not Null

Auction

Attribute	Data type	Nullable
starting_bid	Float	Not Null
min_sale_price	Float	Not Null
getit_now_price	Float	Null
auction_length	Integer	Not Null
auction_end_time	DateTime	Not Null

Category

Attribute	Data type	Nullable
category_name	String	Not Null

Condition

Attribute	Data type	Nullable
condition_name	String	Not Null

Item_Rating

Attribute	Data type	Nullable

stars	Integer	Not Null
comment	String	Null
rate_date_time	DateTime	Not Null

Bid

Attribute	Data type	Nullable
bid_amount	Float	Not Null
time_of_bid	DateTime	Not Null

Auction_Result

Attribute	Data type	Nullable
sale_price	Float	Null
auction_ended	DateTime	Not Null

Auction_Cancelled

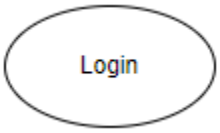
Attribute	Data type	Nullable
cancellation_date_time	DateTime	Not Null
cancellation_reason	String	Null

Business Logic Constraints:

- Users new to BuzzBid must first register.
- Users who have an existing BuzzBid account will not be able to register.
- Only administrative users can add or change the list of categories in the database and cancel the auction.
- To successfully list an item, the **StartingBid** should be lower than **MinSalePrice**, and Minimum Sale Price should be lower than **GetItNowPrice** (if listed).
  (StartingBid < MinSale Price < GetItNowPrice)
- When the auction is active, only the user who listed an item can edit its **Description**.
- Users cannot buy their own items.
- To successfully enter a bid, a user must enter a **Bid Amount** that is at least $1 more than the current highest bid and less than the Get It Now Price.
- The user can immediately buy the item if a Get It Now Price has been set, with a button allowing the user to make the purchase. This immediately ends the auction (even if the time is not yet expired) and sets the current user as the Winner of the auction.
- Only the Winners of an auction can enter a **rating** for an item listed once. An admin or the winner of the item can delete the rating.

Task Decomposition with Abstract Code:

Login



Lock Types: Read-only on User entity

Number of Locks: Single

Enabling Conditions: None

Frequency: Around 250 logins per day

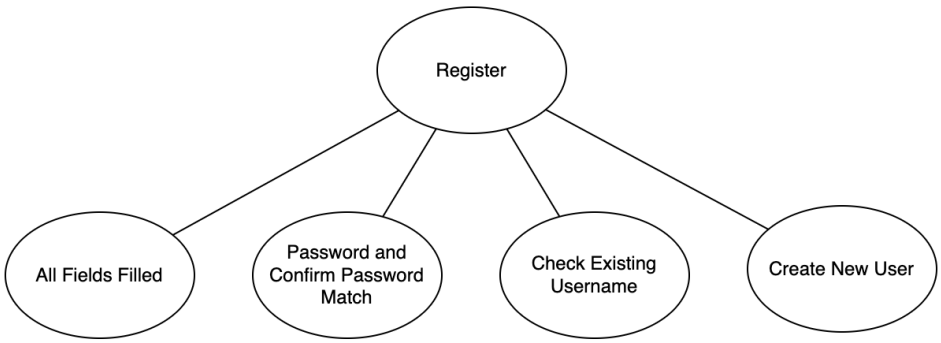
Consistency (ACID): not critical, order is not critical.

Subtasks: Mother Task is not needed. No decomposition needed.

Abstract Code

- User enters username ('$Username), password ('$Password') input fields.
- If data validation is successful for both username and password input fields, then:
    - When ***Enter*** button is clicked:
        - If User record is found but user.password != '$Password':
            - Go back to the **Login** form, with an error message.
        - Else:
            - Store login information as session variable '$Username.
            - Go to **Main Menu** form.
- Else email and password input fields are invalid, display **Login** form, with error message.

Register



Lock Types: Read and Write both on User entity

Number of Locks: Single lock on User entity

Enabling Conditions: New users click on the ***Register*** button

Frequency: Approx. 10 per day

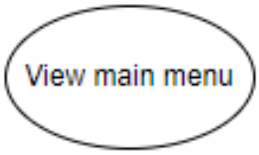
Consistency (ACID): Order is critical.

Subtasks: All tasks must be done. Mother task required. Order is important. Subtasks to check **All Fields Filled** and **Password and Confirm Password Match** should be done first as they don't need any database read or write. **Check Existing Username** should be done after as it needs a database read. Finally the **Create New User** subtask for database write should be done last, only if the other subtasks are successful.

Abstract Code

- User must fill the fields. (*FirstName*, *LastName*, *Username*, *Password*, *Confirm Password*)
- When user clicks on ***Register*** button.
    - If all the fields are not filled:
        - Mention an error msg/ warning that all the fields are required.
    - If Username is already being used:
        - Mention an error msg/ warning that the username is already taken.
    - If the password and confirm password don't match:
        - Mention an error msg/ warning that they should match
    - If all other subtasks are successful, create a new record in the User entity table

View main menu



Lock Types: Read-only on User entity

Number of Locks: Single lock on User entity

Enabling Conditions: After user successfully completes **Login**

Frequency: Same as User Login.

Consistency (ACID): Not critical. Order is not critical.

Subtasks: Mother task not needed. Decomposition not needed.

Abstract Code

- After a user logs in, the firstname and lastname should be listed on the main menu.
- Based on the user_type:
  - If the user is not a administrative user:
    - Show only "**Search for Items**", "**List an Item for Sale**", "**View Auction**" tabs.
  - Else:
    - Along with "**Search for Items**", "**List an Item for Sale**", "**View Auction**" tabs. Display report tabs "**Category Report**", "**User Report**", "**Top Rated Items**", "**Auction Statistics**", "**Cancelled Auction Details**".
- Upon:
  - Click ***Search for Items*** button - Jump to the **Item Search Query** task.
  - Click ***List an Item for Sale*** button - Jump to the **List Item to Sell** task.
  - Click ***View Auction*** button - Jump to the **Auction Results** task.
  - Click ***Category Report*** button - Jump to the **Category Report** task.
  - Click ***User Report*** button - Jump to the **User Report** task.
  - Click ***Top Rated Items*** button - Jump to the **Top Rated Items** task.
  - Click ***Auction Statistics*** button - Jump to the **Auction Statistics** task.
  - Click ***Cancelled Auction Details*** button - Jump to the **Cancelled Auction Details** task.

List Item to Sell

Lock Types: Read lock on Category, Condition entities and Write lock on Item entity

Number of Locks: Multiple locks needed on Item, Category, Condition entities and Lists, Belongs relationships.

Enabling Conditions: User clicks on ***List My Item*** button and the business constraints on Starting Bid, Minimum Sale Price, Get It Now Price are met.

Frequency: Approx 50. per day

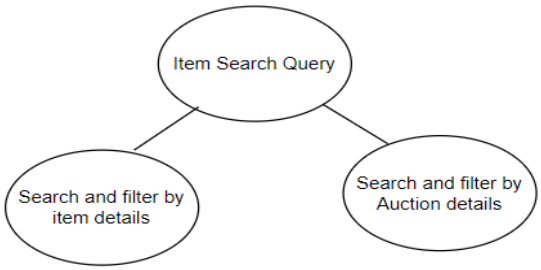
Consistency (ACID): Order is critical.

Subtasks: Mother task needed. Order is important. The database read for **Get Category List** happens before the database write for **List Item in Auction**.

Abstract Code

- Read the database records from the Category entity
  - Display *Category* field dropdown values
- Read the database records from the Condition entity
  - Display *Condition* field dropdown values
- User inputs values for fields *Item Name, Description, Category, Condition, Start auction bidding at (*StartingBid*), Minimum sale price (*MinSalePrice), *Auction ends in (*AuctionLength), *Get it now price, Returns Accepted (*Returnable)
- Check for business constraint StartingBid < MinSalePrice < GetItNowPrice:
  - If true: Create new records for Item and Auction entity
  - If false: Display error message to correct input values

Item Search Query



Lock Types: None

Number of Locks:None

Enabling Conditions: User clicks on search option and enters different search parameters like item name keyword, selects a category from dropdown and minimum/maximum prices and preferred condition of the item. Not all fields are mandatory.

Frequency: Approx 300 per day. (assuming each logged in user searches atleast once)

Consistency (ACID): Not critical. Order is not critical.

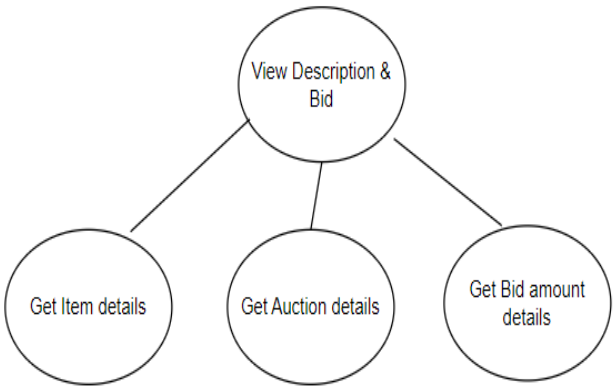
Subtasks: Mother task is not needed. Order is not important. After the user enters the search criteria and hits the Search button, only the items matching all the criteria must be returned. If no search criteria is specified, all the items must be returned.

Abstract Code

- User enters search criteria in the **Item Search** form and hits ***Search*** button
- If Keyword is present, filter items whose item_name or description contains the keywords in anywhere and proceed to the next step. If no keyword is present i.e empty, select all the items and proceed to the next step.
- If Category is selected from dropdown, filter items whose Category matches the selection and proceed to the next step. If no selection is made in the dropdown, select all the items and proceed to the next step.
- If Minimum price is present,
    - if there are bids on the item
        - calculate max bid for the item
        - filter items whose max_bid is greater than or equal to minimum price in search criteria
    - else, filter items whose starting_price is greater than or equal to minimum price in search criteria
- If Maximum price is present,
    - if there are bids on the item

        - skip items whose starting_price is greater than the maximum price in search criteria
        - calculate max bid for the item
        - filter items whose max_bid is less than or equal to maximum price in search criteria
    - else, filter items whose starting_price is less than or equal to minimum price in search criteria
- If Condition is selected from the dropdown, filter items whose Condition matches the selection or better than the selection(New>Very good>Good>Fair>Poor). If no selection is made in the dropdown, select all the items and return.
- The **Search Results** returned should contain the item_id, item_name, current_bid, user_name of the highest bidder, the item's get_it_now_price and the auction_end_time of the item.
- The items should be displayed based on the Auction end date with the auction ending the soonest listed first.
- when the user clicks on the item name, it should take to the item description page which is described below in the "**View Description & Bids**" task.

View Description & Bids



Lock Types: Read-only on Item, Category, User,Bid and Auction entities

Number of Locks: Multiple locks needed on Item, Category, User,Bid , Auction entities and their relationships

Enabling Conditions: When user clicks on item name on the **Item Search Results** form.

Frequency: Approx >300 per day. (assuming each logged in user searches atleast once)

Consistency (ACID): Order is critical.

Subtasks: Mother task needed. Order is important.Fetch item details and bid details on the item and return

Abstract Code

- Get item details and add to result the following fields - item_id, item_name, description, is_returns_accepted
- Get category and condition of the item and add it to result
- Get bid details such as bid_amount, bid_datetime, and username and add it to result
- Add current user's bid amount to your_bid and add it to result
- Get get_it_now_price from Auction if present and add it to result
- ***Edit description*** button should be enabled only for the user who listed the item .
- ***Cancel item*** button should be enabled only for the admin users.
- Return result

Edit Description

Lock Types: Read-only on Category, User, Auction and Read-Write on Item entities

Number of Locks: Multiple locks needed on Item, Category, User, Auction entities and their relationships

Enabling Conditions: In **Item description** form, the user who listed the item clicks on the ***edit description*** button.

Frequency: Low

Consistency (ACID): is not critical, even if the description is being read by other users while editing.

Subtasks: Mother task is needed. Order is important. The item and bid/auction details are read and then the edit description action is performed which updates the item description for an item.

Abstract Code

- Upon clicking the ***edit description*** button, display a pop-up window to enter the new description. Thereby, the existing description is replaced with the new one.

Get It Now

Lock Types: Read-only on Category, Item, User;Bid, Auction and Read-Write on Auction Results

Number of Locks: Multiple locks needed on Item, Category,Bid, User,Auction,Auction Results entities and their relationships

Enabling Conditions: The user clicks on ***Get It Now*** button (user should not be the one who listed this item).

Frequency: Approx 5-10 per day.

Consistency (ACID): Order is critical.

Subtasks: Mother task is needed. Order is important. The item and bid/auction details are read. Update the Auction Results .

Fetch item details and bid details on the item and return; check that the user viewing is not the one who listed the item; After the user clicks on ***get it now***, the user will purchase the item, and this will immediately ends the auction (even if the time is not yet expired) and sets the current user as the Winner of the auction. Then the **auction result** is returned.

Abstract Code

- Fetch Item, User;Bid, Auction details for the item.
- If the user hits ***get it now*** button
  - Check if the auction is active for the item.
  - The user should be able to purchase this item and the auction ends. The auction ended is then changed to this time.
- Auction Results are updated.

Enter Bid

Lock Types: Read-only on Category, Item, User,Auction; and Read-Write on Bid

Number of Locks: Multiple locks needed on Item, Category, User, Auction,Bid entities and their relationships

Enabling Conditions: After the user clicks on ***Bid on this item*** button, if the user is not bidding on their own item, the user will be allowed to enter the bid. Bids must be at least one dollar higher than

the current (highest) Bid Amount. The bid must also be less than the Get It Now Price. If the user attempts to bid higher than the Get It Now Price, an error message should be shown suggesting that the user click ***Get it Now*** instead of bidding. If the bid entered is within allowed range, the **auction** page will be returned.

Frequency: Approx 30-50 per day.

Consistency (ACID): Order is not critical.

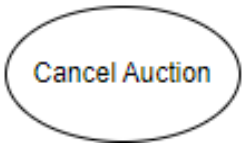
Subtasks: Mother task is needed. Order is important. The item and bid/auction details are read. The user bids on the item.

Fetch **item** details and **bid** details on the item and return; Check if the user viewing the item is the one who listed the item; If the bid entered is within allowed range, the **auction** page will be returned.

Abstract Code

- When a user is viewing an item through **view descriptions and bids** form.
- The minimum bid is displayed on the screen.
- The user will bid on the item.
- The current bid> = minimum bid and current bid<= Get It Now Price
- The user clicks on ***Bid on this Item*** button
- Error message is displayed if the condition is not met for the current bid.
- Bid entity is updated

Cancel Auction



Lock Types: Read-only on Category, User,Item,Auction , Auction Results and write on Auction Cancelled entities

Number of Locks: Multiple locks needed on Item, Category, User, Auction, Auction Results, Auction Cancelled entities and their relationships

Enabling Conditions: The user viewing the **Item description** form should be an admin and click on the ***cancel this Item*** button.

Frequency: Low

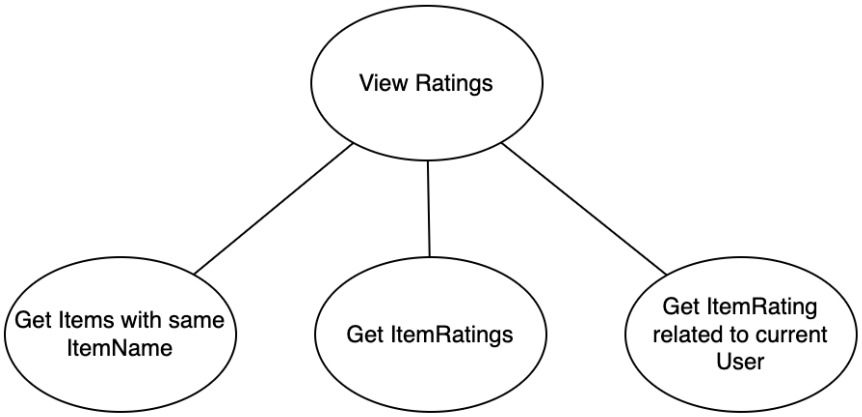
Consistency (ACID): Order is important. As when the auction is being cancelled the user should not bid on the item.

Subtasks: Mother task is needed. Order is important. The item and bid/auction details are read and then the auction is cancelled for an item.

Abstract Code

- The admin can cancel the auction by themselves or by request from the regular user who listed the item.
- The user should be prompted to provide the cancellation reason (which is optional). The date and time of cancellation will be recorded as the cancelled date time in Auction Cancelled .
- This item will be moved to the Auction Results(with sale price and auction ended) and will not be available to user's to bid anymore.

View Ratings



Lock Types: Read lock on ItemRating and Item entities

Number of Locks: Multiple read locks on ItemRating and Item entities

Enabling Conditions: User clicks on ***ViewRatings*** on **Item for Sale** page.

Frequency: Approx 100 times/day

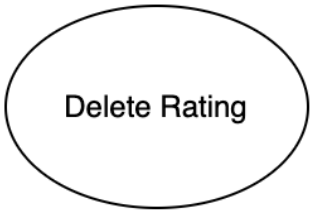
Consistency (ACID): Getting updated data is not critical, parallel edits to records is allowed. Order of subtasks is critical

Subtasks: Mother task needed. Order of subtasks is important.

Abstract Code

- Get the ItemName of the current item for which the user clicks ***ViewRatings***
- Get all Items with the same ItemName and their related ItemRatings
    - Display *Stars, Comment, RateDateTime* for each ItemRating
- Check if the current logged in user has an ItemRating related to the same item
    - Display the *Stars, Comment, RateDateTime* under the ***Delete My Rating*** section

Delete Ratings



Lock Types: No locks needed

Number of Locks: None

Enabling Conditions: User clicks on ***Delete My Rating*** on **Item Ratings** page.

Frequency: Approx 1 per day

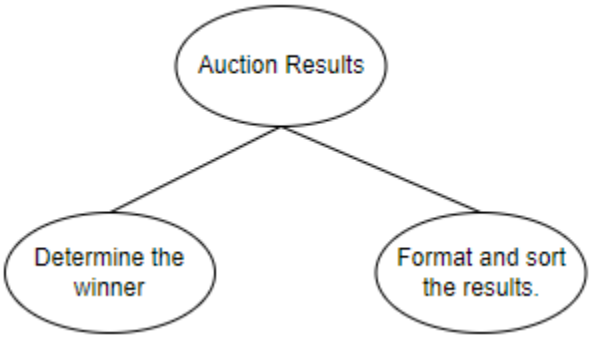
Consistency (ACID): Not critical

Subtasks: Mother task not needed.

Abstract Code

- If the business constraint is met that current logged in user added the rating or admin user:
  - Show the ***Delete My Rating*** button
- User clicks on ***Delete My Rating*** button
- Remove the record from ItemRating
- Refresh the screen to display the **ItemRatings** page again by running the **View Ratings** task again

Auction Results



Lock Types: Read-only on Category, Item, Condition, User,Auction, Auction Results, Auction Cancelled entities.

Number of Locks: Multiple locks needed on Item, Category, Condition, User,Auction, Auction Results, Auction Cancelled entities and their relationships

Enabling Conditions: After the auction is ended, the user can click on ***Auction Results*** button to see the auction results.

Frequency: Approx 5-10 per day.

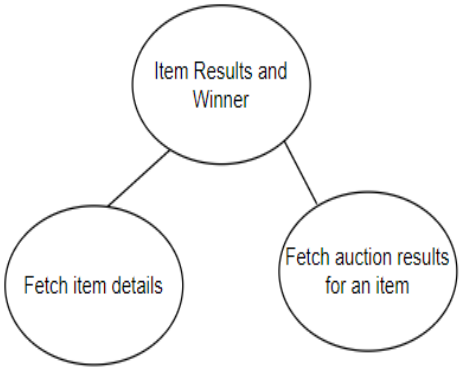
Consistency (ACID): Not critical.

Subtasks: Mother task is needed. Order is important.

Abstract Code

- If the ***Get it now*** button is hit or the auction time has expired for an item
  - If the item was purchased using the ***Get It Now*** option
    - Set SalePrice as the Get It Now Price and the winner as the Username of the person who purchased the item.
  - If the high Bid >= Minimum Sale Price,
    - Set SalePrice as the value of the highest Bid and the winner as the Username of the person with the highest Bid
  - If no one purchased the item using the ***Get It Now*** option, and if the highest Bid < Minimum Sale Price, or there are no Bids at all
    - Both the sale price field and the winner field will be left blank.
  - If the auction was cancelled:
    - SalePrice will be left blank, the winner will be listed as "Cancelled", and the cancellation date/time is listed
- The above validation is done to determine the auction results when the ***View auction result*** button is clicked from the **View Main Menu** task.
- Show the latest auction results sorted by latest auction ended data/time with ItemID, SalePrice, Winner, Auction End DateTime

Item Results & Winners



Lock Types: Read-only on Item, Category, User,Bid,Auction Results, Auction Cancelled entities

Number of Locks: Multiple locks needed on Item, Category, User,Bid , Auction, Auction Results, Auction Cancelled entities and their relationships

Enabling Conditions: When user clicks on item name from the **Auction Results** form.

Frequency: Approx 5-10 per day.

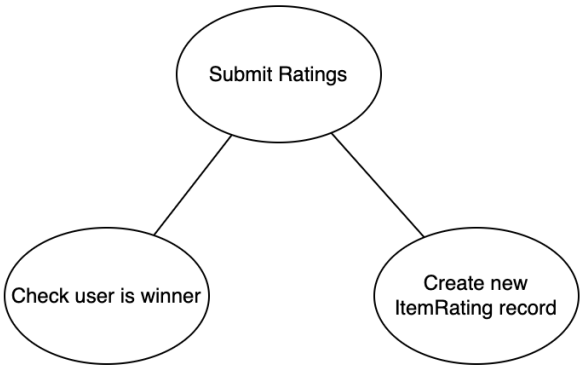
Consistency (ACID): Order is critical.

Subtasks: Mother task needed. Order is important.Fetch item details and bid details on the item and return

Abstract Code

- Get item details and add to result the following fields - *item_id, item_name, description, is_returns_accepted*
- Get *category* and *condition* of the item and add it to result
- Get *get_it_now_price* from Auction if present and add it to result
- Get *Auction ended* from Auction Results and add it to the result.
- Get bid details such as *bid_amount, bid_datetime*, and *username* and add it to result
- Sort the bids by *Bid Amount* (descending)
- If the item has a winner then highlight the first bid row to green.
- If the item has no winner then highlight the first bid row to yellow.
- If the item auction is cancelled then highlight the first bid row to red and display the username as "Administrator".
- Return result

Submit Ratings



Lock Types: Write lock not needed as business constraint ensures only one user will be able to create the rating for the item (no concurrent edits possible)

Number of Locks: Not needed

Enabling Conditions: User is the winner or the item auction and user clicks on ***View Ratings*** button from the **Item Results** page

Frequency: Approx 5 per day

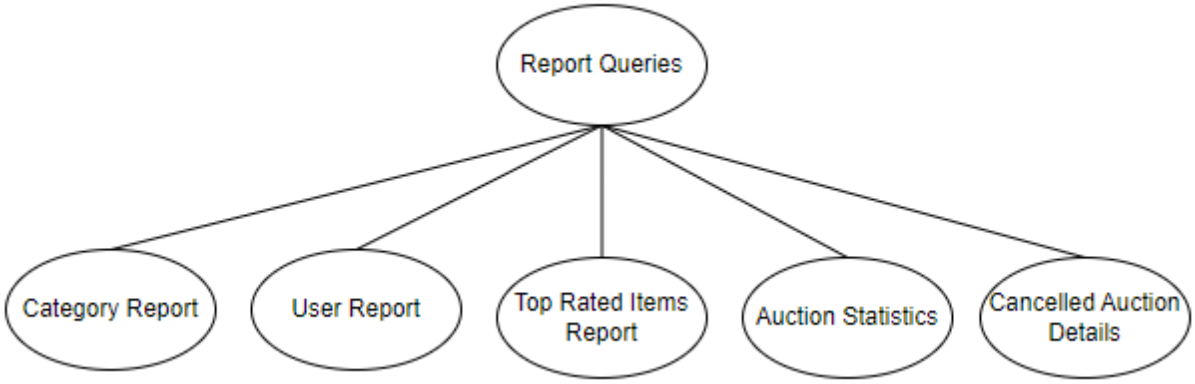
Consistency (ACID): Order is critical.

Subtasks: Mother task is needed. Order of subtasks is important to check the business constraint before.

Abstract Code

- If the business constraint is met that current logged in user is the winner of the item auction:
    - Display the *My Rating (*Stars) and *Comments* on the **ItemRatings** page at the bottom.
- User enters input and clicks on the ***Rate This Item*** button
- Create a new record for ItemRating entity and rates relationship with User and has relationship with Item
- UI screen should refresh by running the **View Ratings** task again so that the recently added rating is shown under in under the ***Delete My Rating*** section

Report Queries



- ## Category Report

Lock Types: Read-only on Category,Item,Auction,Auction Results entities

Number of Locks: Multiple locks needed on Category, Item, Condition, Auction, Auction Results entities and their relationships

Enabling Conditions: The admin user clicks on the ***Category report*** button from the **view main menu** form.

Frequency: Approx 20-25 per day

Consistency (ACID): Not critical. Order is not needed.

Subtasks: No mother task needed.

Abstract Code

- Read from Category,Item,Auction,Auction Results.
- Aggregate information for each Category:

    - The minimum Get It Now Price, the maximum Get It Now Price, the average Get It Now Price across all items in the category is taken as the Min Price, Max Price and Average Price.
    - Total number of items in the category.
    - If the Get it Now Price is not available they should be not be part of Min Price, Max Price and Average Price but included in the item count.
    - The items from the cancelled auctions should not be considered for the aggregation.
- Sort the aggregated information by category name in alphabetical order.
- The fields in the report are Category, Total Items, Min Price, Max Price, Average Price.

User Report

Lock Types: Read-only on Rating, User, Item, Condition, Auction, Auction Results entities

Number of Locks: Multiple locks needed on Category, User, Item, Condition, Auction Results entities and their relationships

Enabling Conditions: The admin user clicks on the ***User report*** button from the **view main menu** form.

Frequency: Approx 20-25 per day

Consistency (ACID): Not critical. Order is not needed.

Subtasks: No mother task needed.

Abstract Code

- Read from Rating, User, Item, Condition, Auction, Auction Results entities.
- Aggregate information about users:
    - Total number of items listed by the user (whether sold or not)
    - Total number of items sold by the user (either via Get It Now or auction)
    - Total number of items won by the user (either via Get It Now or auction)
    - Total number of items that the user rated
    - if the user has listings:
        - the most frequent item condition for the user's listings (in the event of a tie, the lowest-quality condition should be used).
    - If the user has no listings:
        - this field should display "N/A".
- Return report with UserName and aggregated information

Top Rated Items Report

Lock Types: Read lock on ItemRating and Item entities

Number of Locks: Multiple read locks on ItemRating and Item entities

Enabling Conditions: An AdminUser clicks on the ***Top Rated Items*** button from **Main Menu** page

Frequency: Approx 20-25 per day

Consistency (ACID): Not critical

Subtasks: No mother task needed.

Abstract Code

- Read *Stars* from Item Ratings and *ItemName* from Item entities
- Calculate Average Rating of all the *Stars* from ItemRatings related to each Item with same *ItemName*
- Filter Top 10 records by Average Rating and display their:
  - *ItemName* attribute from Item entity
  - *Average Rating* (derived earlier)
  - *Rating Count* that is the number of ItemRatings entities associated to each Item with same *ItemName*

• Auction Statistics

Lock Types: Read-only on Item, Ratings, Auction, Auction_Result and Auction_Cancelled entities

Number of Locks: Multiple locks needed on Item, Ratings, Auction, Auction_Result and Auction_Cancelled entities and their relationships

Enabling Conditions: The admin user clicks on the ***Auction Statistics*** report button from the **view main menu** form.

Frequency: Approx 20-25 per day

Consistency (ACID): Not critical

Subtasks: No mother task needed.

Abstract Code

- Read from Auction, Auction_Result and Auction_Cancelled entities.
- Aggregate information about auctions:
  - Total number of active auctions (auction_end_time is past current time)
  - Total number of finished auctions
    - current time is past auction_end_time AND
    - auction was not canceled
  - Total number of auctions with winners (either via Get It Now or auction)
  - Total number of canceled auctions
- Aggregate information about ratings:
  - Total number of items with ratings
  - Total number of items without ratings
- Return report with auction and ratings report details.

- ## Canceled Auction Details

Lock Types: Read-only on Item, User and Auction_Cancelled entities

Number of Locks: Multiple locks needed on Item, User and Auction_Cancelled entities and their relationships

Enabling Conditions: The admin user clicks on the ***Canceled auction*** report button from the **view main menu** form.

Frequency: Approx 20-25 per day

Consistency (ACID): Not critical

Subtasks: No mother task needed.

Abstract Code

- Read from Item, User and Auction_Cancelled entities.
- Get a list of item ids that were canceled from Auction_Canceled entity and map the below details for each item:
  - Add item id, cancel date time and cancel reason from Auction_Results entity
  - List all the item details from Item entity for the retrieved item ids
  - fetch the username who listed the item and add it to the report result
- Return report with Canceled auction details.